MACHINE LEARNING PROJECT

Telling customers when their order will arrive

Predicting Olist delivery times from information known at checkout

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BUSINESS VALUE

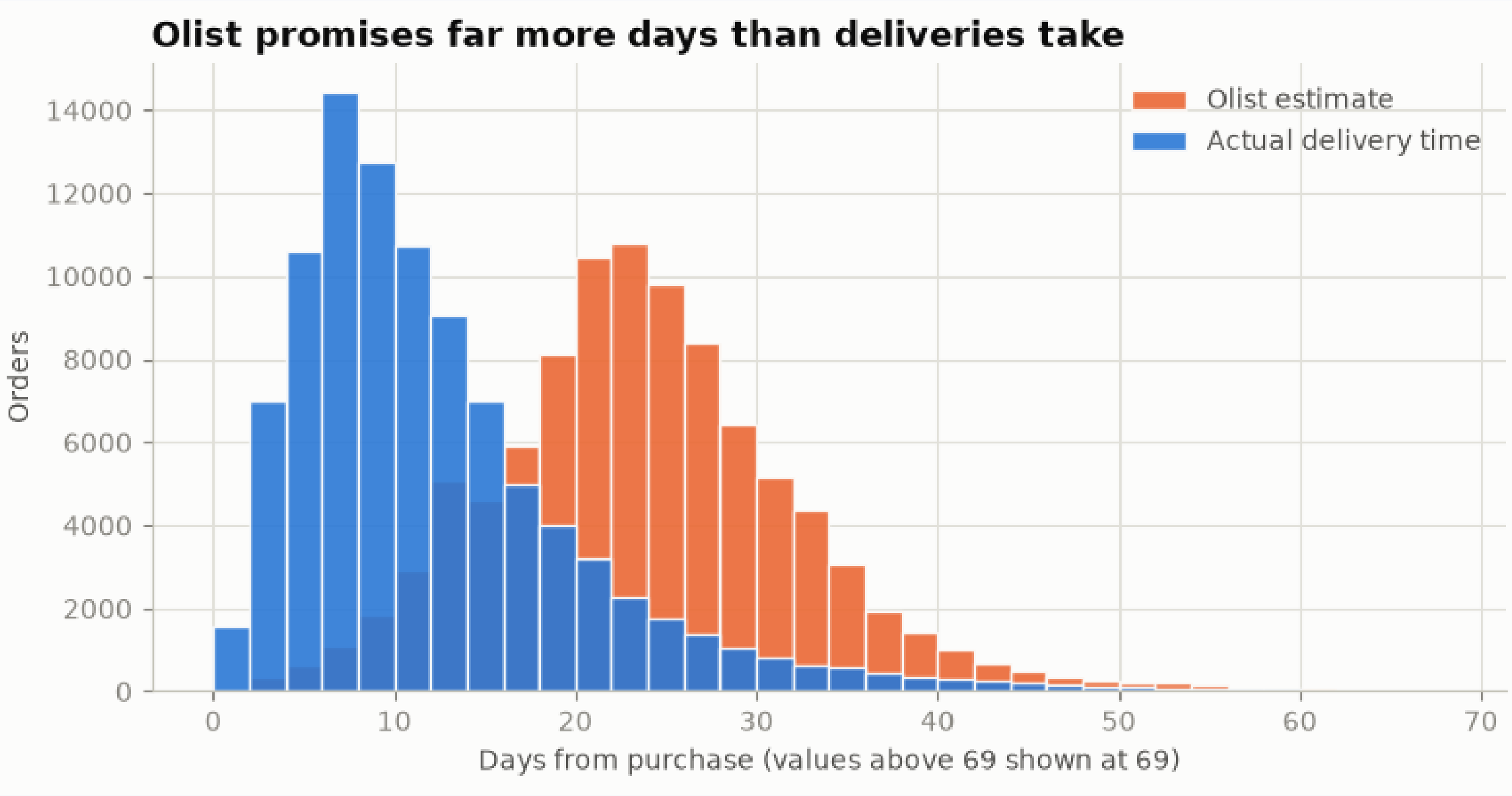
More accurate dates help customers and daily operations

Customers

- Delivery dates they can plan around
- A shorter, honest wait shown before they buy

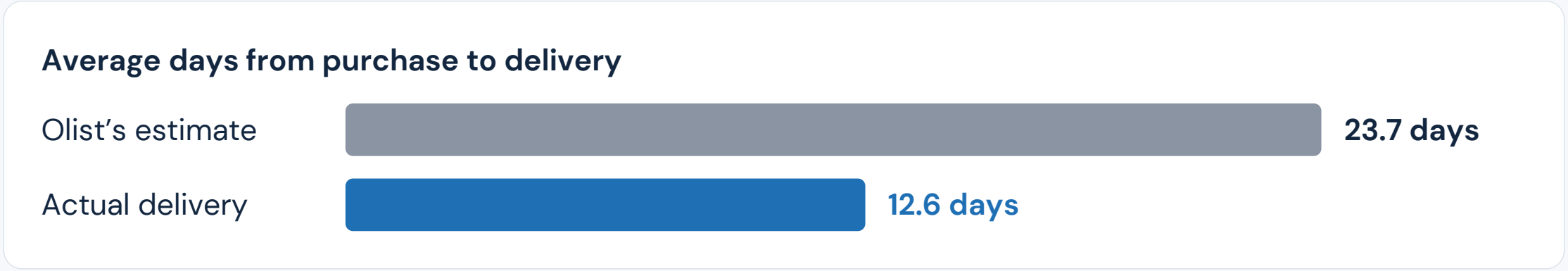
Operations

- Fewer “where is my order?” contacts
- Promises that match what carriers deliver


Olist promises far more days than deliveries take
Olist estimate
Actual delivery time
Orders
14000
12000
10000
8000
6000
4000
2000
0
0
10
20
30
40
50
60
70
Days from purchase (values above 69 shown at 69)

THE PROBLEM

Customers are quoted almost twice the wait they actually have



OUR SOLUTION

Our model is about three times more accurate than today's estimate

12.7 days

average miss of Olist's estimate today

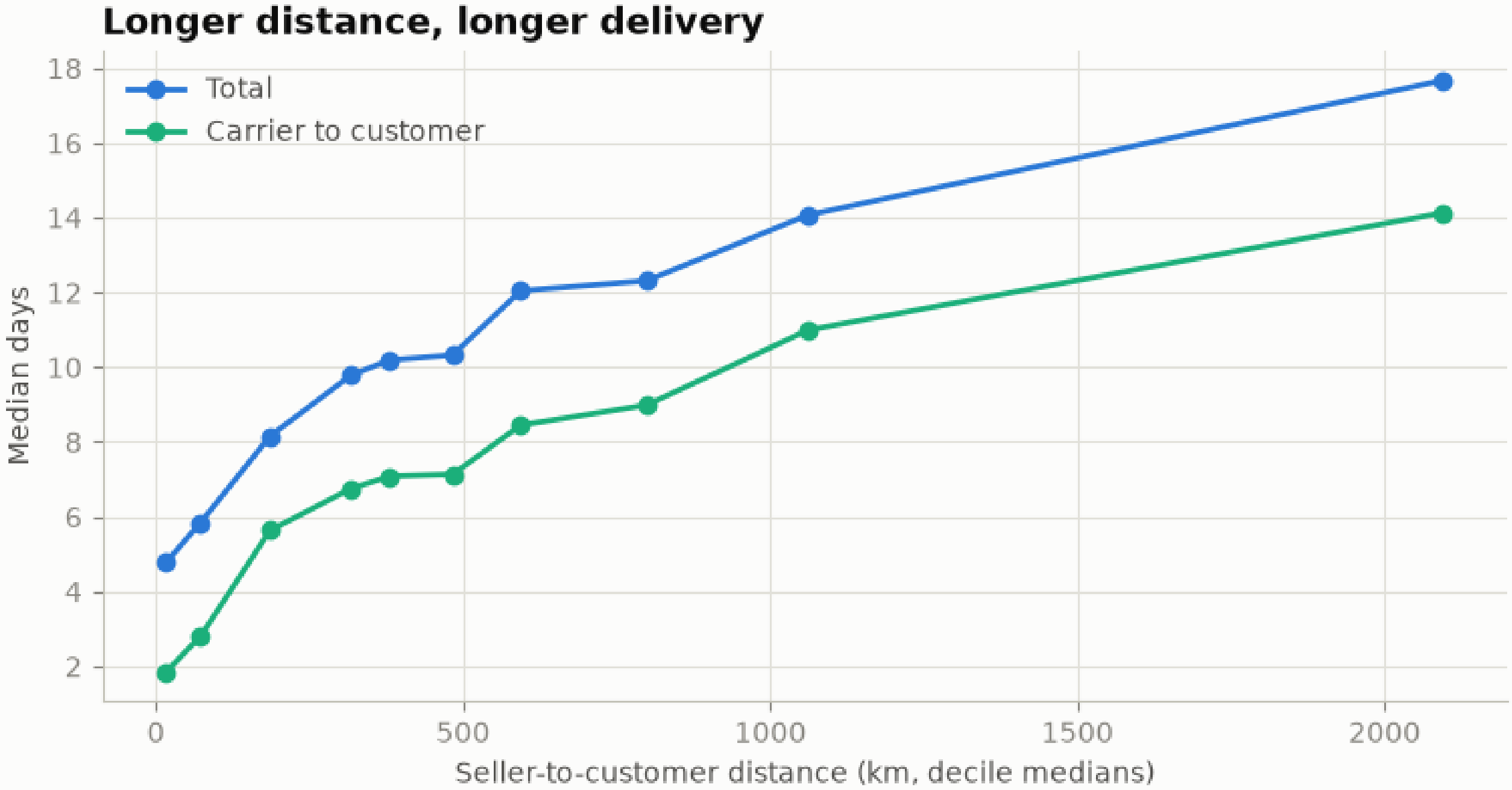
4.3 days

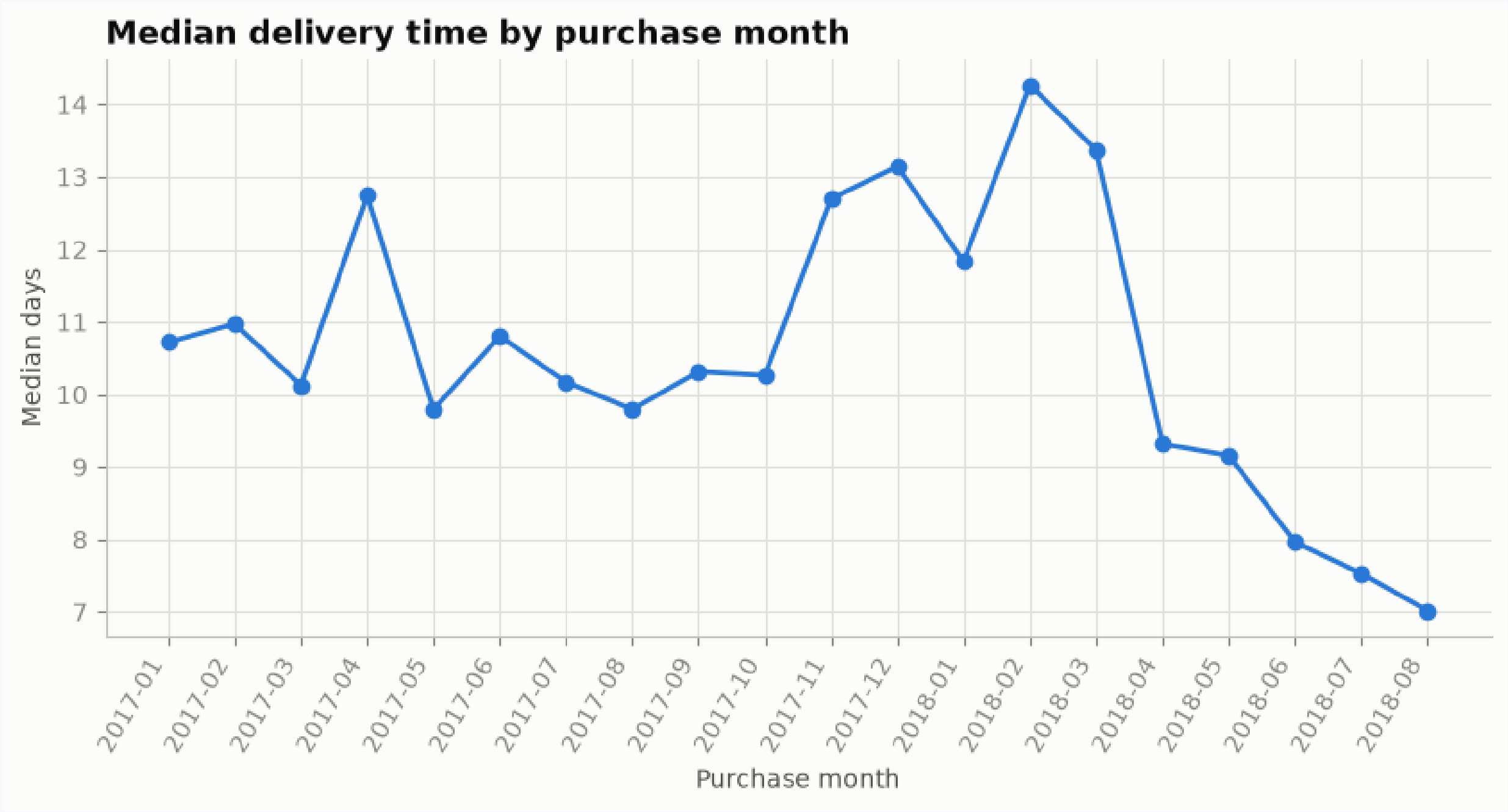
average miss of our model

66%

smaller error

Average error measured on 19,257 orders the model had never seen


Longer distance, longer delivery
Total
Carrier to customer
Median days
18
16
14
12
10
8
6
4
2
0
500
1000
1500
2000
Seller-to-customer distance (km, decile medians)


Median delivery time by purchase month
Median days
14
13
12
11
10
9
8
7
2017-01
2017-02
2017-03
2017-04
2017-05
2017-06
2017-07
2017-08
2017-09
2017-10
2017-11
2017-12
2018-01
2018-02
2018-03
2018-04
2018-05
2018-06
2018-07
2018-08
Purchase month

KEY DATA INSIGHTS

Shipping, geography and season explain most of the waiting time

74%

of the wait is shipping to the customer; the seller's preparation is only 26%

2×

Orders within one state arrive in ~6.6 days; orders across states take ~12.8 days

12–14

days at the Nov 2017 – Mar 2018 peak, against 10–11 days in most of 2017

> Where an order ships from and to matters far more than what is in the box.

Source: 96,281 delivered Olist orders, 2016–2018

DATA AND PREPARATION

From raw orders to a model-ready dataset

99,441
orders in the data

→

96,281
delivered, with complete dates (97%)

→

19,257
orders kept hidden for the final test

What we used

- Route: where it ships from and to, and the distance
- Order: price, shipping fee, weight, size, items
- Timing: month, weekday and hour of purchase

Missing information

- A few orders lack a location or product size
- Gaps are filled with typical values, so no order is lost

Unusual orders

- 64 extreme orders (over 100 days) are left out of learning
- They stay in the final test, so our scores stay honest

Only information known at purchase is used; nothing from after the order is placed.

HOW WE MEASURE SUCCESS

Success is the average miss, in days

MAE

Mean Absolute Error: how many days off the prediction is, on average

- In days, so anyone can read it
- Not distorted by a handful of extreme orders
- Directly comparable with Olist's estimate: 12.7 days today

Illustrative example (days)

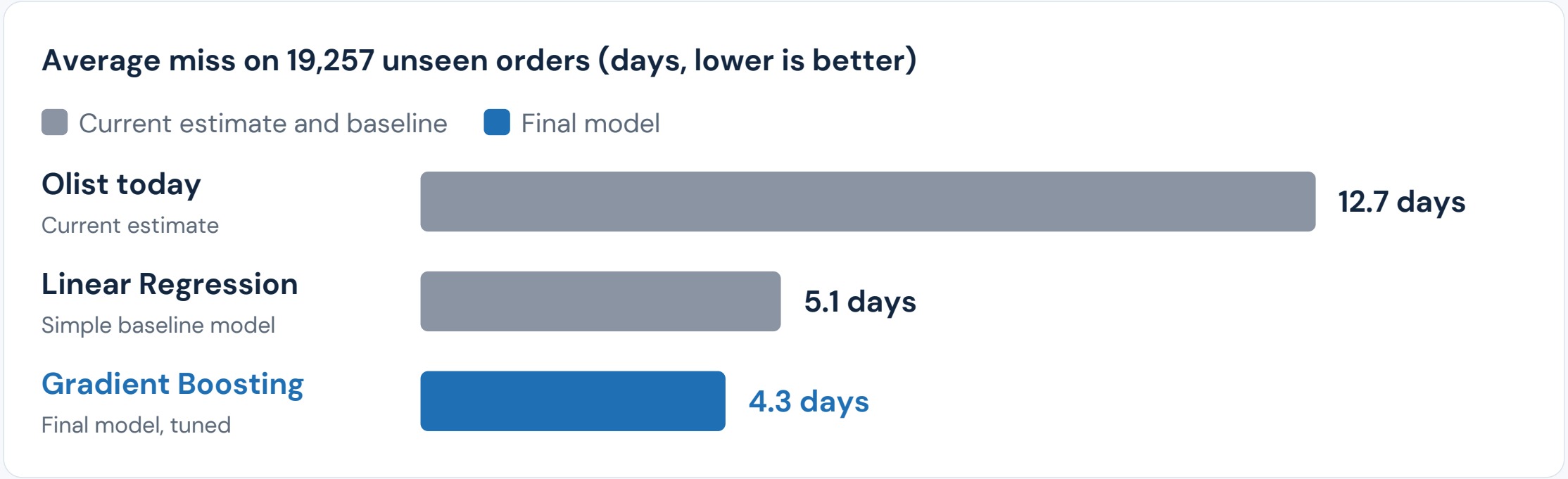
Order	Predicted	Actual	Off by
A	10	14	4
B	12	9	3
C	7	11	4
D	15	14	1

Average miss: (4 + 3 + 4 + 1) ÷ 4 = 3 days

Example numbers are made up to show the calculation; our real results are on the next slide.

MODEL COMPARISON

Tuned model: average miss down to 4.3 days



Test set: 19,257 orders never used for learning or tuning.

MODEL CHOICE

Why we chose Gradient Boosting

Most accurate

4.3 days off, against 5.1 for the simple Linear Regression model.

One direct prediction

Predicting the total time directly (4.3 days) is as good as adding separate handling and shipping predictions (4.35), and simpler.

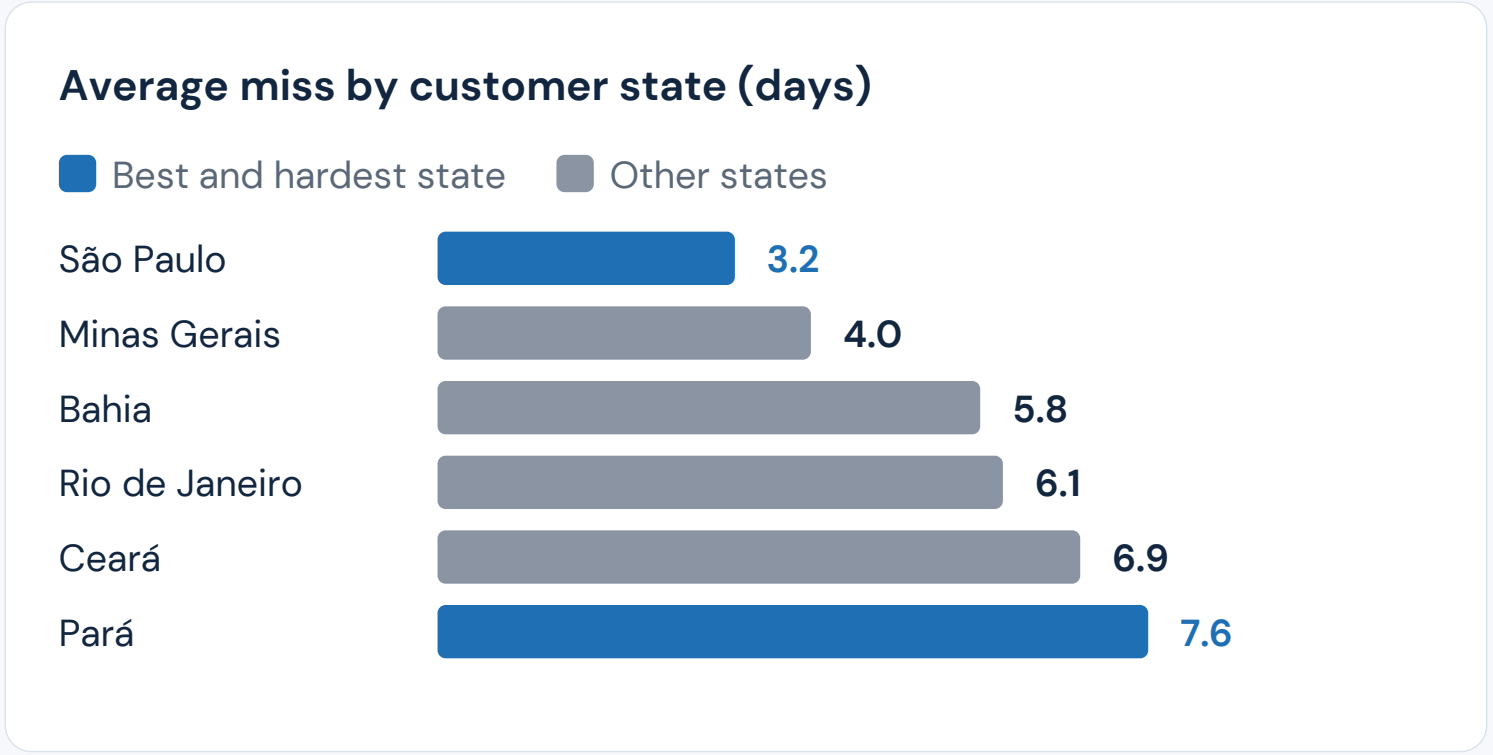
Light enough for checkout

A model file of about 1 MB that runs without special hardware.

Fine-tuning the model's settings added only a small gain: 4.52 to 4.49 days on validation data.

LIMITS

Where the model struggles



2.4×

Remote states such as Pará have more than double São Paulo's error

5–6 days

Average miss in the Nov 2017 – Mar 2018 peak, against 2–3 days in Jun – Aug 2018

Selected states shown. Test set: 19,257 orders.

LIVE DEMO

A customer places an order

1

Choose a product

Pick an item from the store

2

Choose the delivery address

Select the city it should go to

3

Click “Order Now”

Place the order as a customer would

4

See the estimate

The delivery date appears on the confirmation screen

NEXT STEPS

Recommendations and next steps

1

Improvement

Our model cuts the average miss in delivery time from 12.7 to 4.3 days using only information known at checkout.

2

Improve remote routes

Add seller and carrier history, such as a seller’s past handling time, to cut the error where it is highest.

3

Reduce random factors: add outside data

Bring in weather forecasts and holiday calendars. Weather changes can delay carriers, and strikes can too.


Thank you